

# CSE350: Digital Electronics & Pulse Techniques

## Lecture 09: ADC Intro, Flash ADC

Course Instructor:



*MD. ZAYED BIN ZAHIR ARJU (ZBH), ADJUNCT LECTURER  
Department of Computer Science and Engineering (CSE)  
BRAC University*

**Analog:** Information or data that can be any real value at any instant of time. Physical world signals are of analog nature.



Analog Signal

**Digital:** Information in the form of binary bit streams. Computers & all digital devices need digital signals to operate.

Digital data means binary bit of strings.

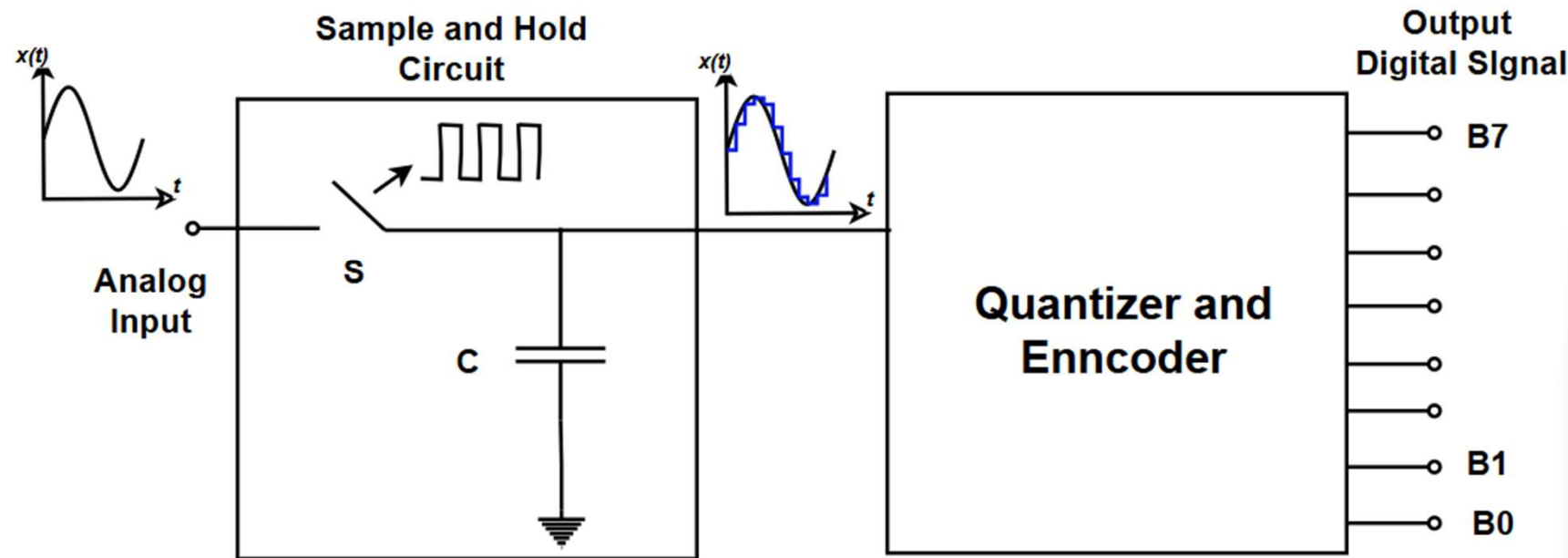


Digital Signal

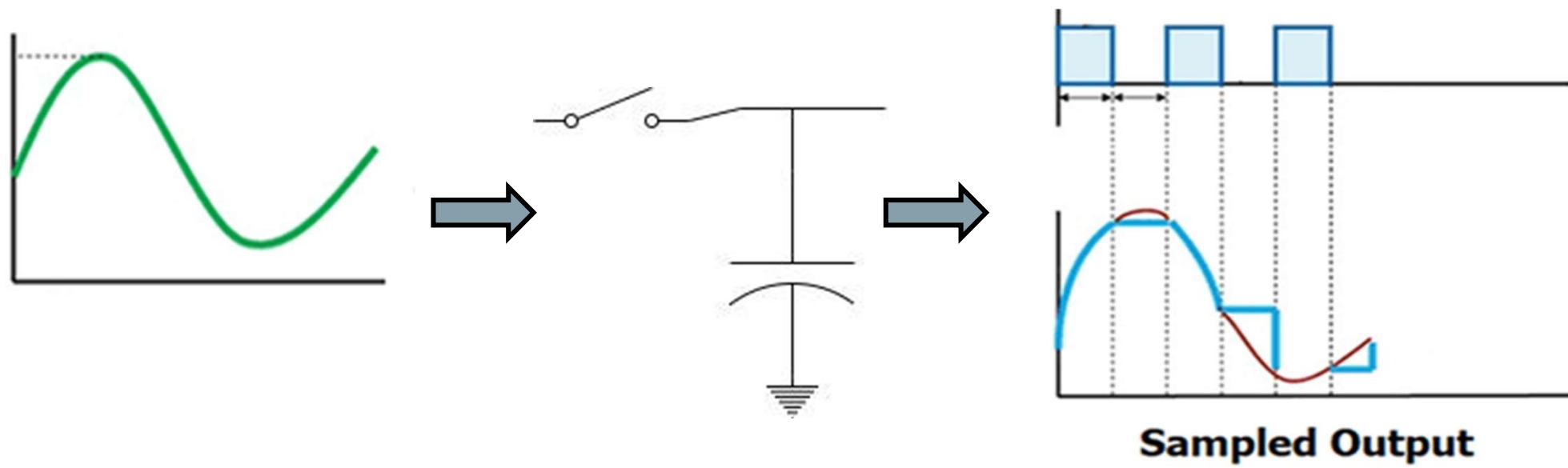
**ADC** is the process of *converting analog signals to digital signals*. It can be summarized as below:



## A/D conversion process



**Sample & Hold:** The analog signal is sampled at regular intervals of a clock pulse. A capacitor is used to hold the peak value of the preceding interval until next one arrives. Thus, instead of all the real values of the original signal, we get a signal that has reduced specific values, i.e. a discrete signal.



# Quantization

**Quantization** is a process by which we assign sampled and hold signal data to some fixed preassigned values.

Depending on the method used to assign analog values to these levels, quantization can be of the following **two** types:

1. Midrise Quantization
2. Midtread Quantization

$N$  = Number of bits

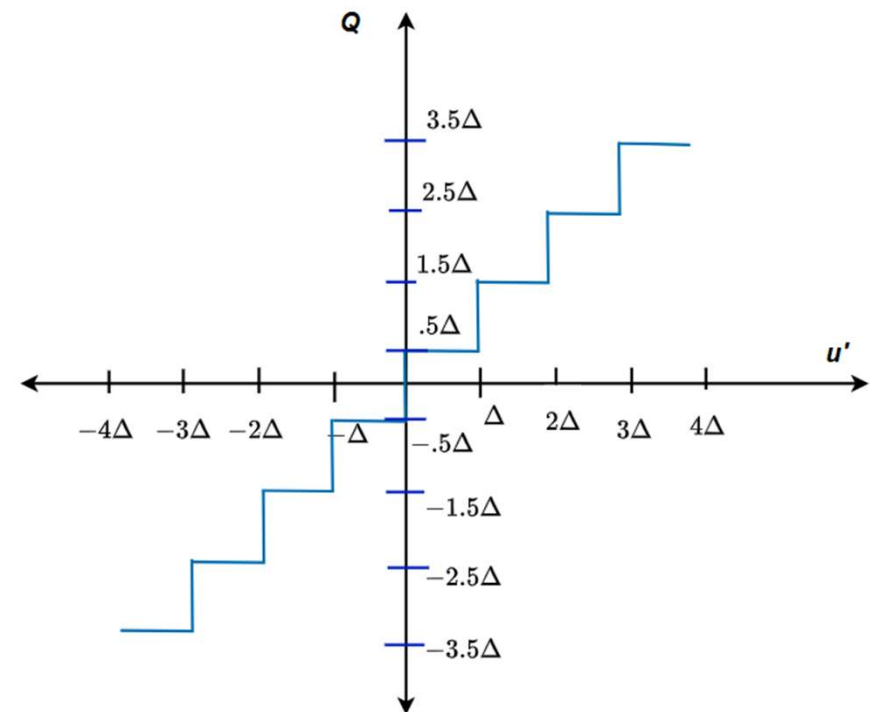
$2^N$  = Number of Quantization levels

The difference between any two consecutive levels are equal and called the **resolution of quantization**. This can be expressed as,  $\Delta = \text{resolution} = \frac{V_{\max} - V_{\min}}{2^N}$

**Mid-rise Quantization:** A type of quantization scheme used in digital signal processing, where the range of the input signal is divided into a set of equal-sized intervals, and the output levels are assigned to the midpoints of these intervals.

The quantization levels are symmetric about zero but **do not include a quantization level at zero (midpoint between minimum & maximum)**. For a signal quantized into 8 levels with a range from -4 to 4:

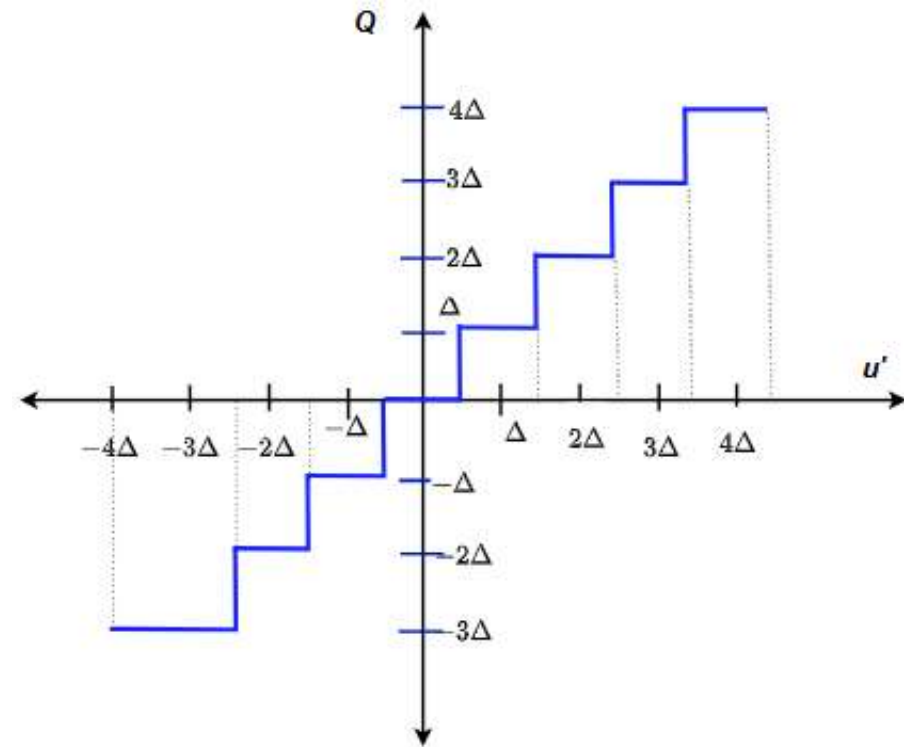
- Quantization intervals:  $[-4, -3), [-3, -2), \dots, [3, 4)$
- Midpoints (output levels):  $-3.5, -2.5, -1.5, -0.5, 0.5, 1.5, 2.5, 3.5$



**Mid-tread Quantization:** A type of quantization scheme used in signal processing, where the range of the input signal is divided into intervals, and the output levels are assigned to the midpoints of these intervals, with one of the levels located exactly at zero (midpoint between minimum & maximum).

For a signal quantized into 8 levels with a range from -4 to 4:

- Quantization intervals:  $[-4, -2.5), [-3.5, -2.5), \dots, [3.5, 4.5)$
- Midpoints (output levels): -3, -2, -1, 0, 1, 2, 3, 4



**Encoding:** This is the process of assigning binary bit streams to the quantization levels.

For example, the **mid-rise quantization** levels as showed in previous slide, will be assigned the following 3-bit values:

**-3.5:** 000 (0)

**0.5:** 100 (4)

**-2.5:** 001 (1)

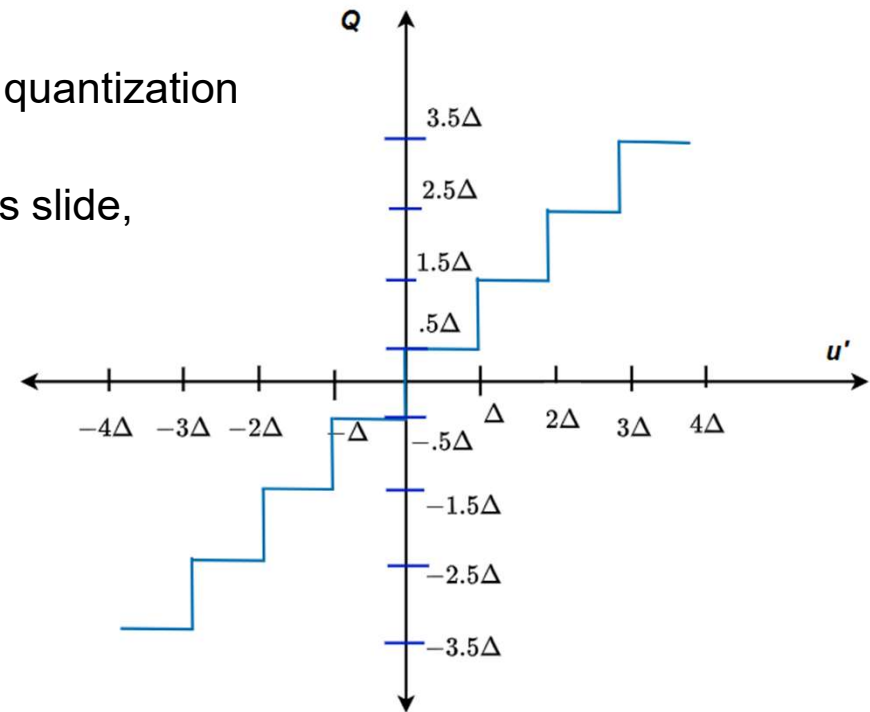
**1.5:** 101 (5)

**-1.5:** 010 (2)

**2.5:** 110 (6)

**-0.5:** 011 (3)

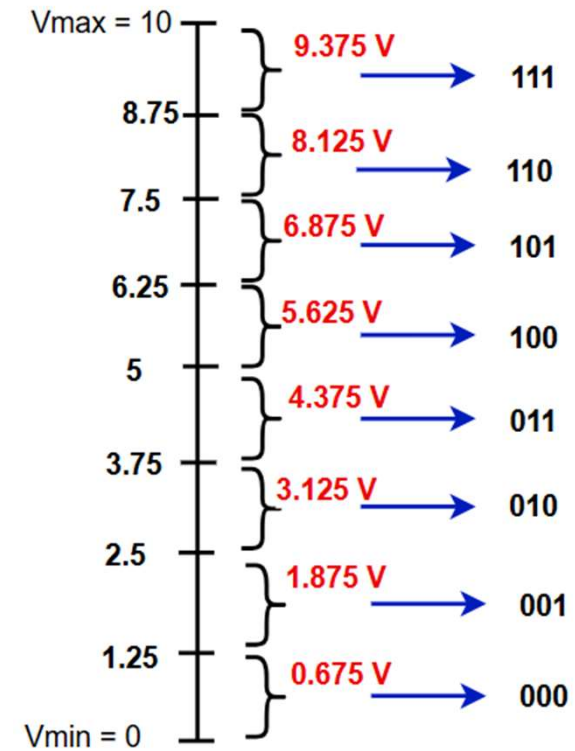
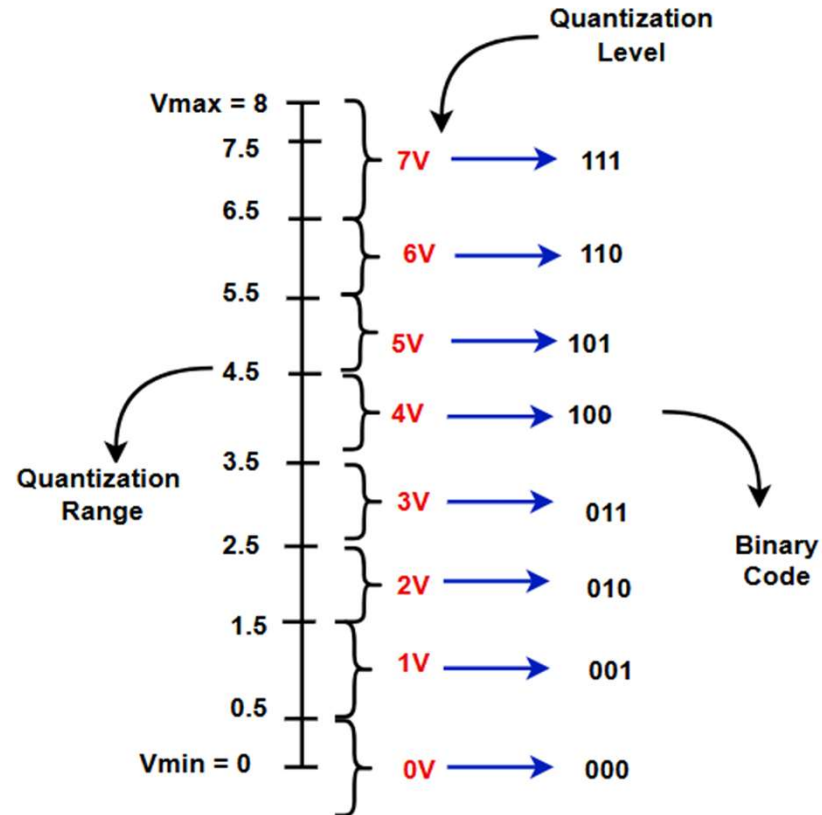
**3.5:** 111 (7)



Example: Identify which one is Mid rise and Mid tread.

1.  $V_{\max} = 8$  and  $V_{\min} = 0$

2.  $V_{\max} = 10$  and  $V_{\min} = 0$



(i) Here,

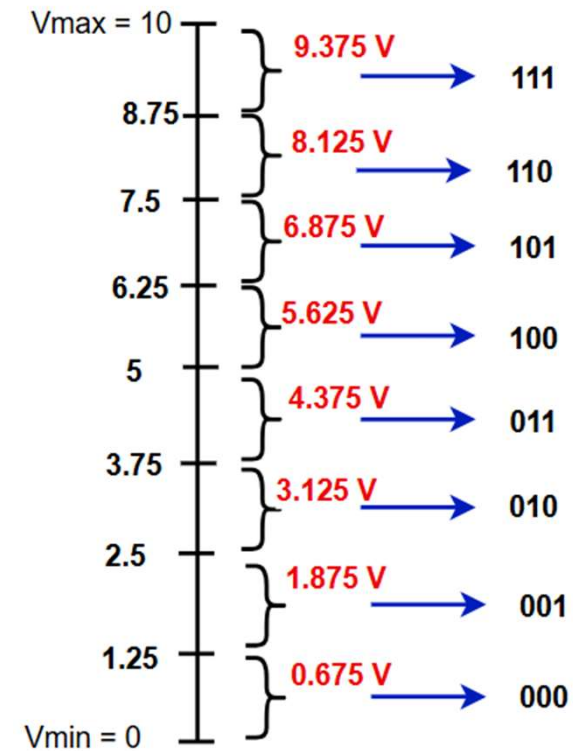
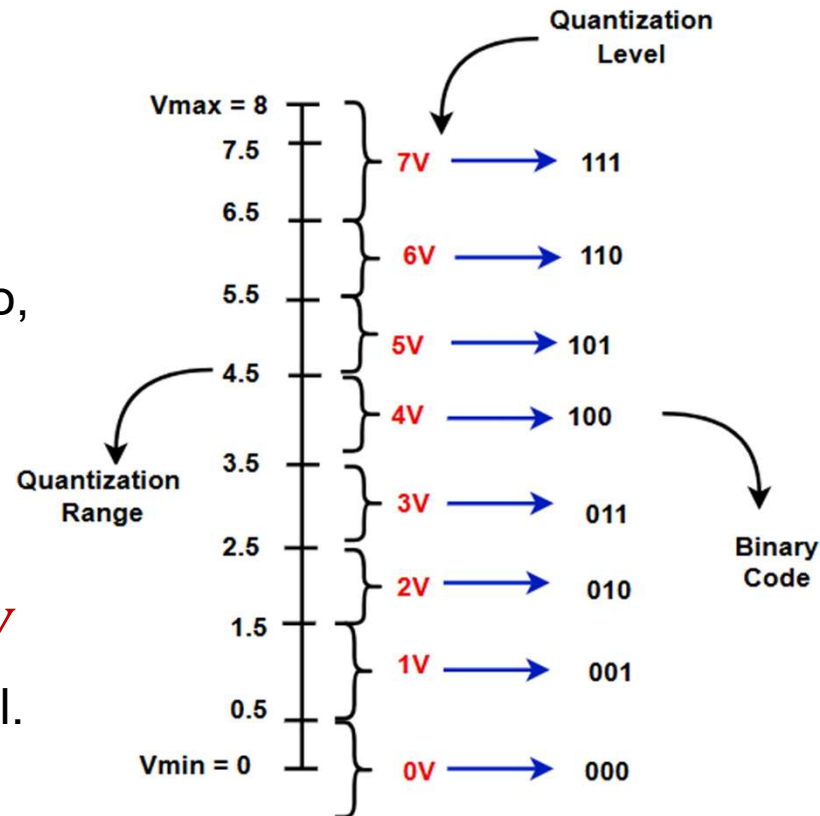
$$\frac{V_{max} + V_{min}}{2} = \frac{8 + 0}{2} = 4 \text{ V}$$

As 4 V is a Quantization level. So, it is a **mid tread quantization**

(ii) Here,

$$\frac{V_{max} + V_{min}}{2} = \frac{10 + 0}{2} = 5 \text{ V}$$

As 5 V is not a Quantization level. So, it is a **mid rise quantization**



# Flash A/D converter

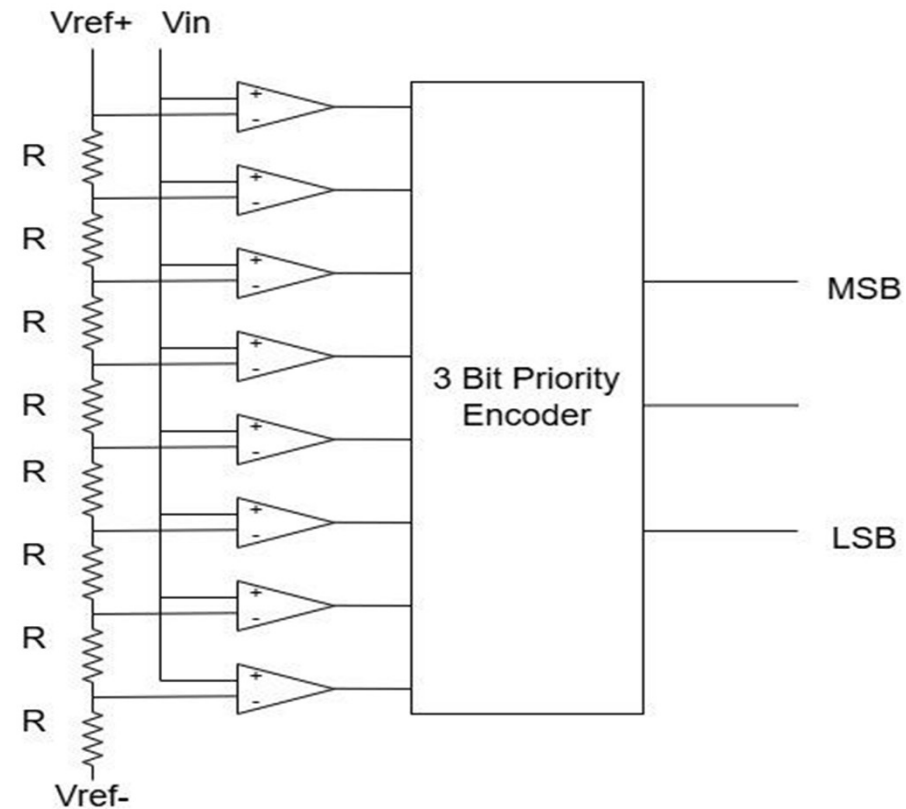
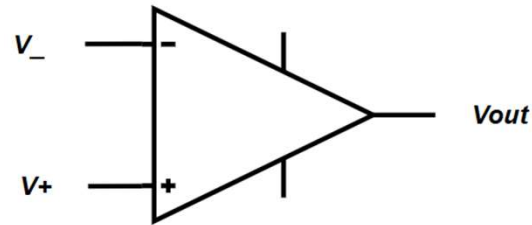
Features of the Flash A/D converter:

- ✓ This converter is very fast. (adv)
- ✓ It requires one clock cycle to convert the analog to digital data. (adv)
- ✓ It requires a lot components. (dis-adv)

We know, For open loop comparator

If  $V_+ > V_-$ ,  $V_{out} = \text{High (logic 1)}$

If  $V_+ < V_-$ ,  $V_{out} = \text{Low (logic 0)}$



## Flash ADC design

Suppose you have a signal with  $V_{\max} = 10\text{ V}$  and  $V_{\min} = 0\text{ V}$ . Design a simple flash ADC converter. Use 3 bits for designing. Provide your encoder truth table.

Design:

Here,

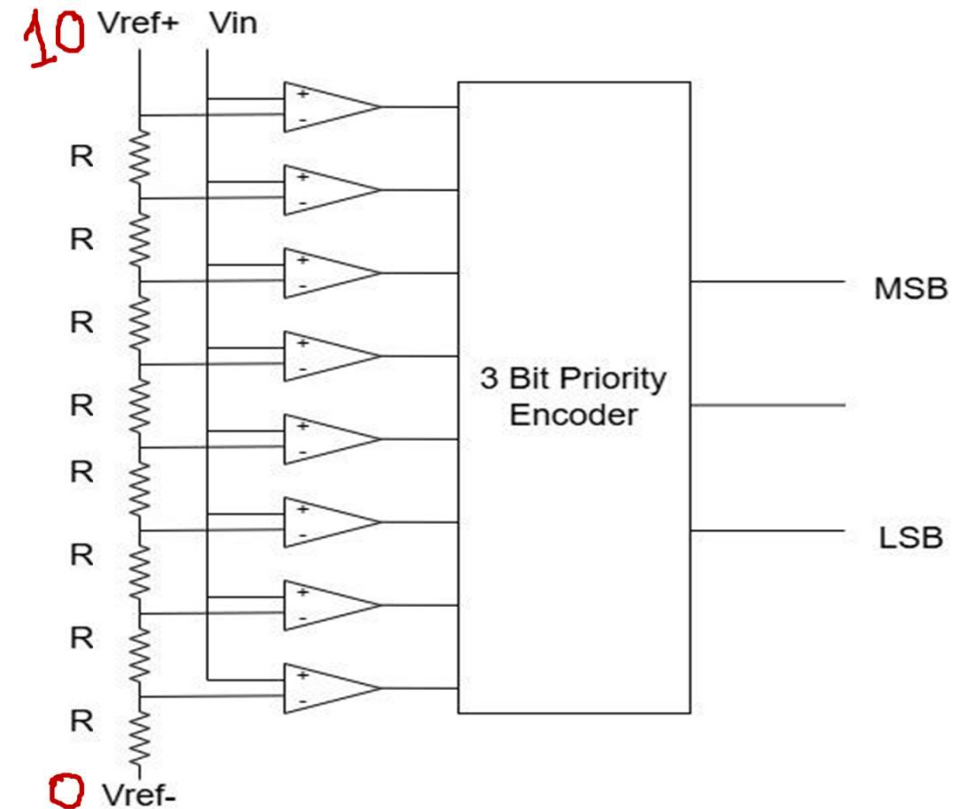
$$N = 3$$

$$\text{\#Number of resistor} = 2^3 = 8$$

$$\text{\#Number of comparator} = 2^3 - 1 = 7$$

$$V_{+ref} = V_{\max} = 10\text{ V}$$

$$V_{-ref} = V_{\min} = 0\text{ V}$$



## Flash ADC design

( V+ of the comparator is connected to the input. )

The following is the Circuit of a 3 bits Flash ADC,

$$I = \frac{10 - 0}{8R} = \frac{10}{8R}$$

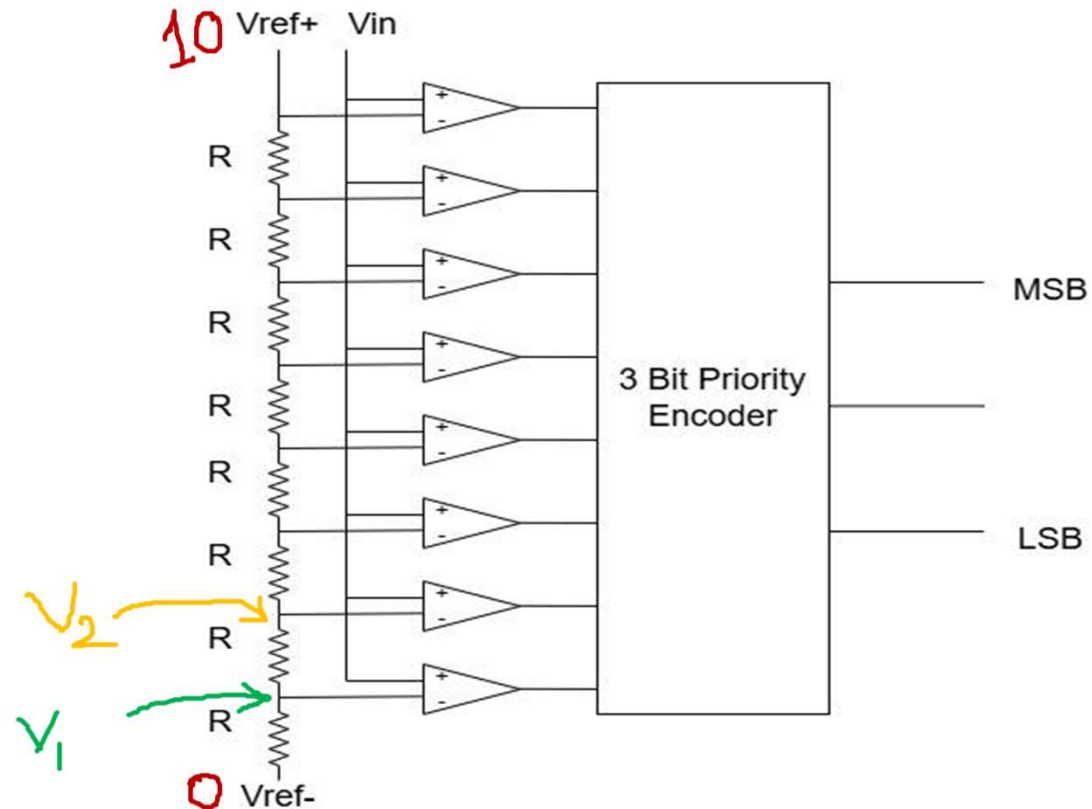
$$I = \frac{V1 - V_{-ref}}{R}$$

$$V1 = V_{-ref} + IR = 0 + \frac{10}{8R} * R = 1.25 V$$

Similarly,  $I = \frac{V2 - V1}{R} \Rightarrow V2 = V1 + IR$

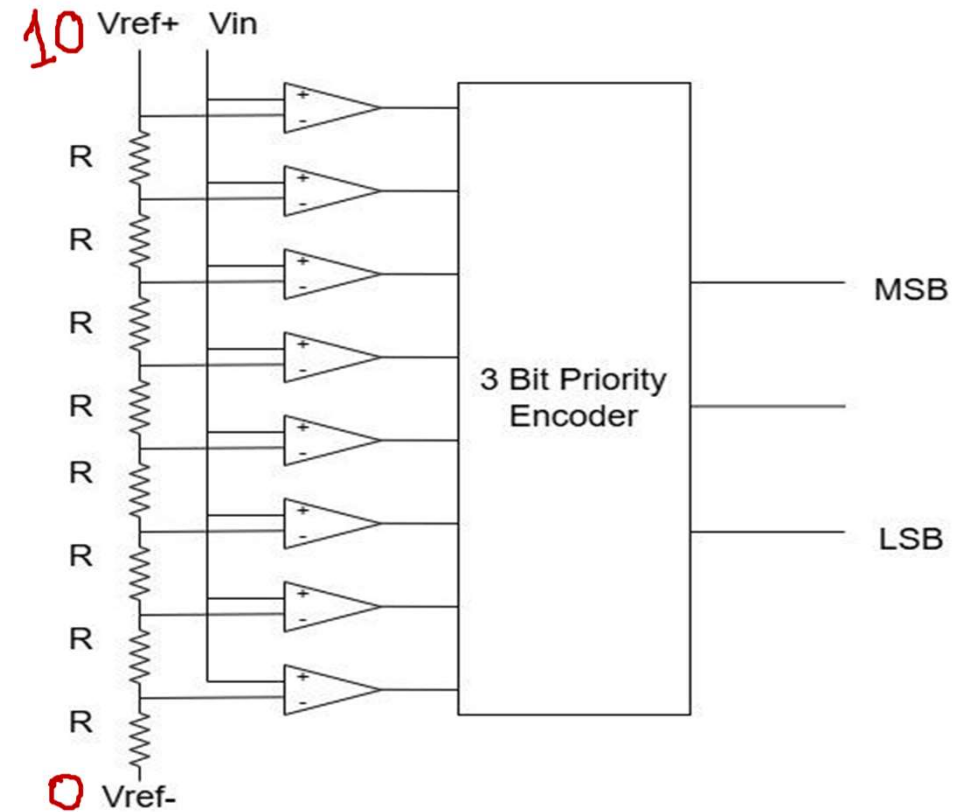
$$\Rightarrow V2 = 1.25 + \frac{10}{8R} * R = 2.5 V$$

You can find other voltage in this way.



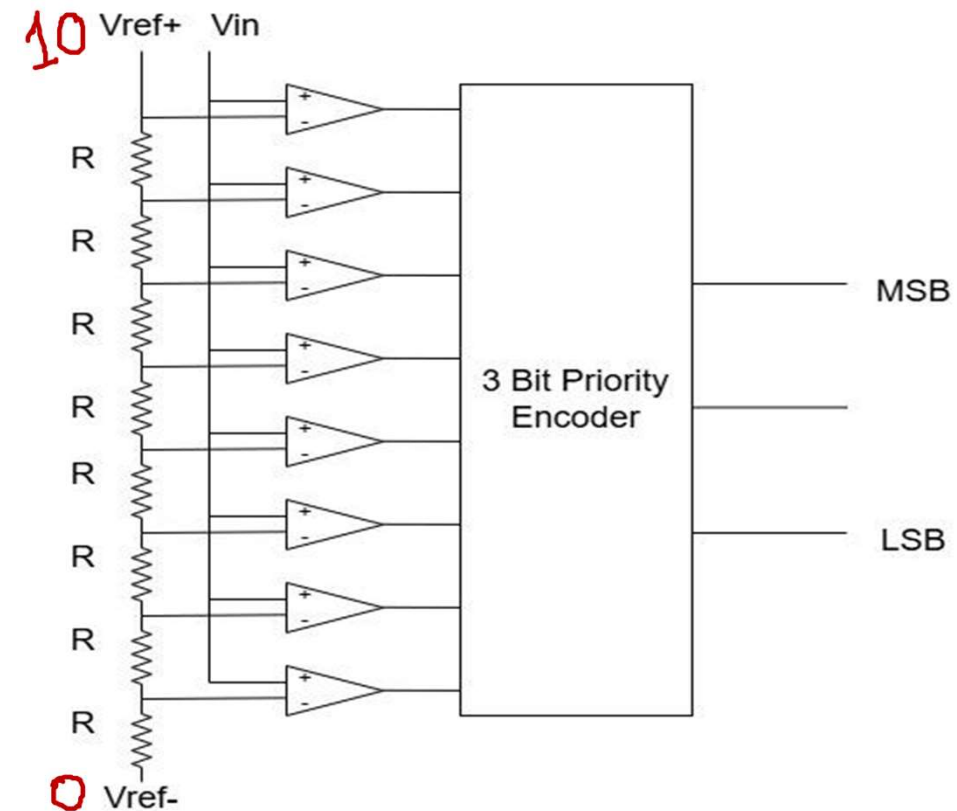
(  $V_+$  of the comparator is connected to the input. )

| Q Range    | Q level | 7 6 5 4 3 2 1 | Binary<br>(B2B1B0) |
|------------|---------|---------------|--------------------|
| 0 - 1.25   | 0.675   | 000 000 0     | 000                |
| 1.25 - 2.5 | 1.875   | 000 000 1     | 001                |
| 2.5 - 3.75 | 3.125   | 000 001 1     | 010                |
| 3.75 - 5   | 4.375   | 000 011 1     | 011                |
| 5 - 6.25   | 5.625   | 000 111 1     | 100                |
| 6.25 - 7.5 | 6.875   | 001 111 1     | 101                |
| 7.5 - 8.75 | 8.125   | 011 111 1     | 110                |
| 8.75 - 10  | 9.375   | 111 111 1     | 111                |



( V- of the comparator is connected to the input. )

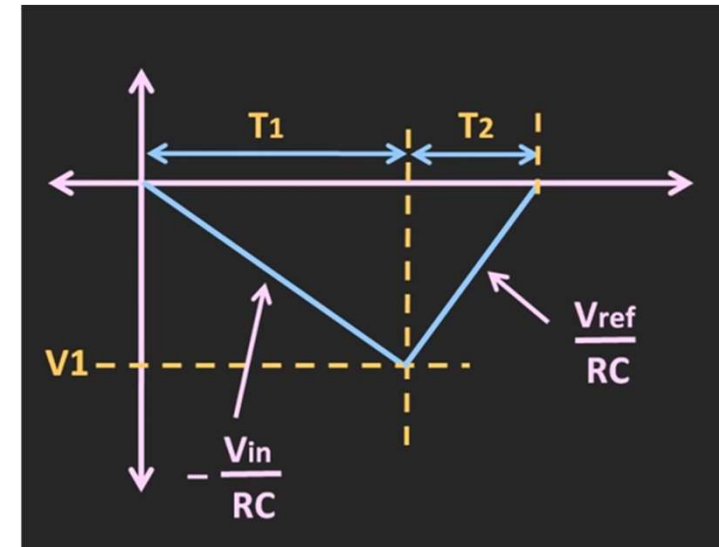
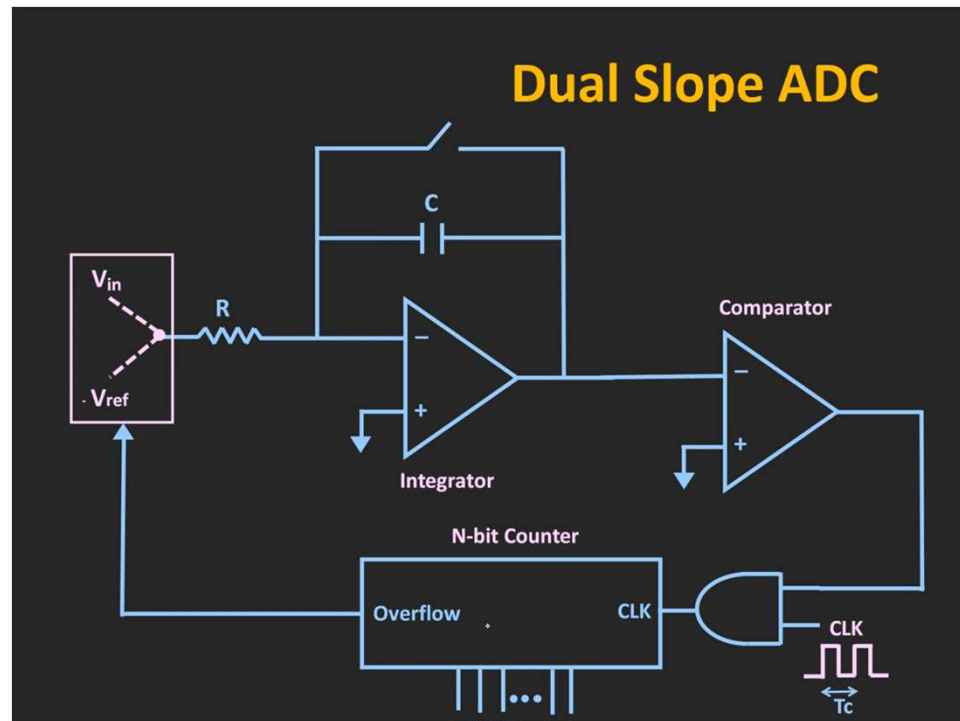
| I7I6I5I4I3I2I1 | Binary<br>(B2B1B0) |
|----------------|--------------------|
| 111 111 1      | 000                |
| 111 111 0      | 001                |
| 111 110 0      | 010                |
| 111 100 0      | 011                |
| 111 000 0      | 100                |
| 110 000 0      | 101                |
| 1 000 000      | 110                |
| 000 000 0      | 111                |



## Dual Slope ADC:

**Advantages:** Requires less components, high resolution (number of quantization levels or bits can be high).

**Disadvantages:** Too slow conversion.

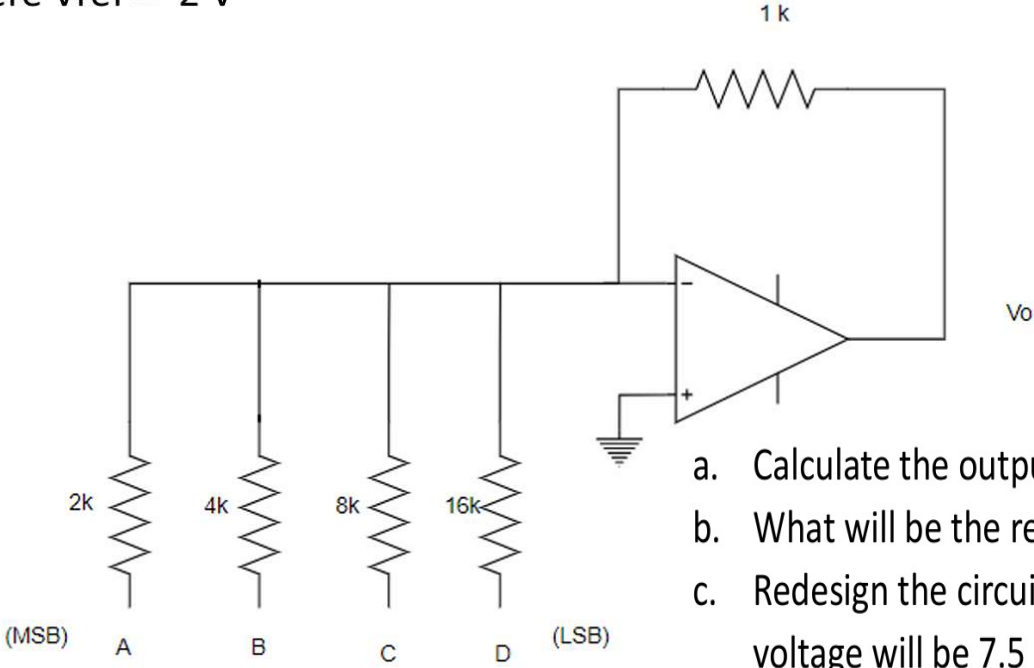


Reference - [https://www.youtube.com/watch?v=2gF\\_nfaBV\\_0](https://www.youtube.com/watch?v=2gF_nfaBV_0)

## Practice Problems – ADC and DAC

5. Design a 4 binary weighted DAC circuit with reference voltage of 5 V and maximum output voltage of 15V.

7. Here  $V_{ref} = -2\text{ V}$



- a. Calculate the output value for 1111 and 1010 input value.
- b. What will be the resolution of this circuit?
- c. Redesign the circuit by changing the value of the feedback resistor so that maximum output voltage will be 7.5 V.

7. a)

$$V_o = -V_{ref} \times R_f \left( \frac{A}{R_A} + \frac{B}{R_B} + \frac{C}{R_C} + \frac{D}{R_D} \right)$$

For, ABCD = 1111

$$V_o = 2 \times 1k \left( \frac{1}{2k} + \frac{1}{4k} + \frac{1}{8k} + \frac{1}{16k} \right)$$

$$= 1.875V$$

7. b) Resolution will be the value for LSB,

$$V_o = 2 \times 1k \left( 0 + 0 + 0 + \frac{1}{16k} \right)$$

$$\text{Resolution} = \frac{1}{8} V \text{ (Ans.)}$$

7. c) For  $V_o \rightarrow \text{Max}$

ABCD = 1111

$$V_o = 7.5 = 2 \times R_f \left( \frac{1}{2k} + \frac{1}{4k} + \frac{1}{8k} + \frac{1}{16k} \right)$$

$$R_f = 4k\Omega$$

1. An analog signal in the range of -5V to 5V is to be converted to a digital signal. Suppose a 2 bit quantization will be implemented.

- a. What is the step size of the ADC circuit?
- b. How many quantization levels are there ?
- c. Find the maximum value of the quantization error.
- d. Calculate the number of resistors and OP-AMP required for the Flash ADC circuit.
- e. Make a table with quantization range, quantization level and encoded binary value.
- f. At some instances the signal values of the ADC circuit are 3.35V, -2.67V and 1.14 V. Find corresponding encoded binary value.
- g. Explain the input and output relationship of the Encoder with a table.

$$1.a) \text{ Step size} = \frac{V_{\max} - V_{\min}}{2^n}$$

$$= \frac{5 + 5}{2^2}$$

$$= 2.5 \text{ V}$$

1.b) Number of quantization level

$$= 2^n$$

$$= 2^2$$

$$= 4$$

1.c) Maximum value of quantization error

$$= \frac{\Delta}{2}$$

$$= \frac{2.5}{2}$$

$$= 1.25 \text{ V}$$

1.d) # of resistor =  $2^n = 2^2 = 4$

# of OP-AMP =  $2^n - 1 = 2^2 - 1 = 3$

1.e) Here,  $\Delta = \frac{5 - (-5)}{2^2} = 2.5V$

| Q. Range  | Q. level | Encoded Value<br>( $b_2 b_1$ ) |
|-----------|----------|--------------------------------|
| -5 — -2.5 | -3.75    | 00                             |
| -2.5 — 0  | -1.25    | 01                             |
| 0 — 2.5   | +1.25    | 10                             |
| 2.5 — 5   | +3.75    | 11                             |

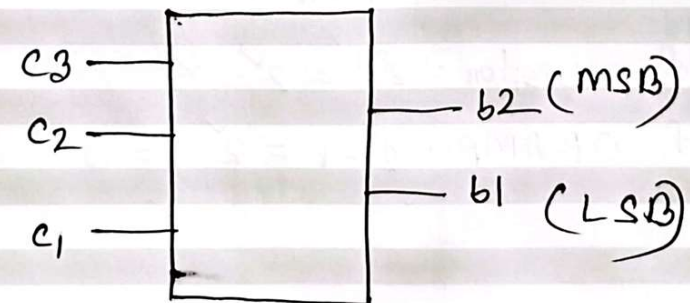
1.f) From previous table (1.e)

if,  $V_{sig} = 3.35V$ , encoded binary ( $b_2 b_1$ ) = 11

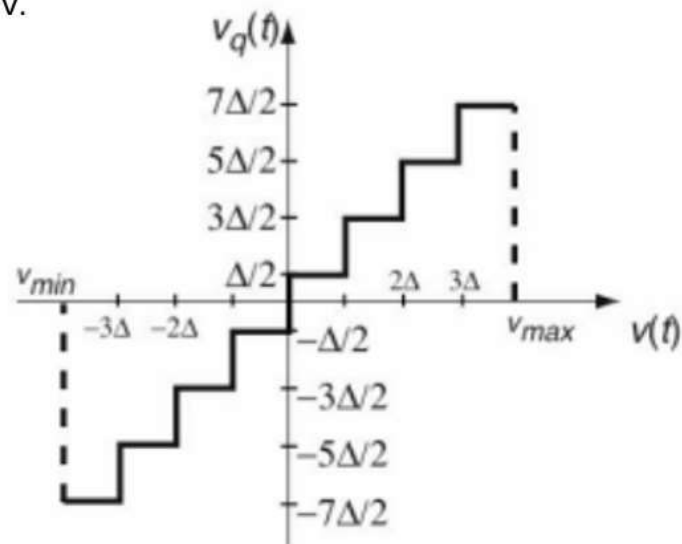
$V_{sig} = -2.67V$ , " " " = 00

$V_{sig} = 1.14V$ , " " " = 10

1.g) Encoder required,



4. The following plot is the relationship between input and output of a midrise quantizer. Where the step size is  $2V$ .



midrise quantizer

- What is the number of binary bits required to express the quantization levels?
- Find the corresponding quantized value if the inputs of the quantizer are  $-5.6\text{ V}$  and  $7.34\text{ V}$ .
- What will be the maximum value of the quantization error?
- Design a full ADC circuit consisting of maximum current of the circuit should not exceed  $0.1\text{ mA}$ .

4.a) There are 8 quantization level in y axis

$$2^n = 8$$

$$n = 3$$

4.b) Given,  $\Delta = 2V$

$$v(t) = -5.6V$$

This value will ~~be~~ fall in the range,

-3 $\Delta$  to -2 $\Delta$  or -6V to -4V range

$$V_q = -2.5\Delta = -2.5 \times 2 = -5V$$

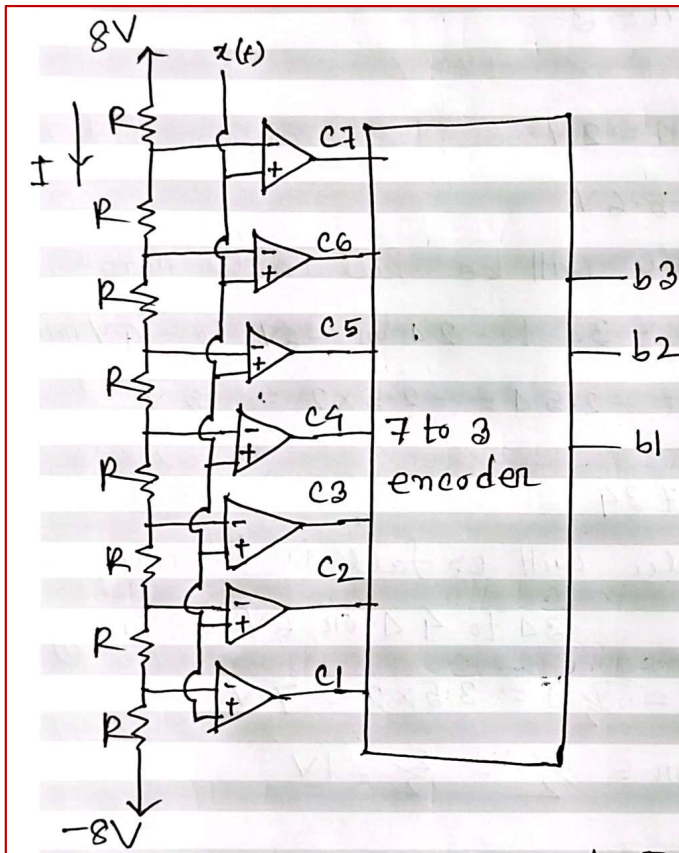
$$v(t) = 7.34$$

This value will ~~be~~ fall in the range,  
3 $\Delta$  to 4 $\Delta$  or 6V to 8V range.

$$V_q = \frac{7}{2}\Delta = 3.5 \times 2 = 7V$$

$$4.c) \text{ Max error} = \frac{\Delta}{2} = \frac{2}{2} = 1V$$

$$4.d) \# \text{ bits} = 3$$



$$\text{resistor} = 2^3 = 8$$

$$\text{OP-AMP} = 2^3 - 1 = 7$$

Given,  $I = 0.1 \text{ mA}$ , and,  $I = \frac{8 - (-8)}{8R} = \frac{2}{R}$

$$\therefore \frac{2}{R} = 0.1 \text{ m} \Rightarrow R = 20 \text{ k}\Omega$$